

Maternal Immune Activation and Schizophrenia:

Investigating alterations in reward anticipation

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Abstract

Negative symptoms are considered a core aspect of schizophrenia, and are largely untreated by current antipsychotic medication. Negative symptoms contain two domains: diminished expressivity and diminished motivation, with the latter incorporating deficits in anticipatory pleasure. Exposure to maternal immune activation is a developmental risk factor in the later onset of schizophrenia; the maternal immune activation model induces immune activation in pregnant animals allowing for the assessment of its role in the development of schizophrenic-like outcomes in offspring. Pregnant Sprague-Dawley rats were injected with LPS or saline at one of three gestational periods (GD10-11, or 15-16, or 18-19). After reaching adulthood, male offspring were tested on measures of reward prediction. It was found the offspring exposed to LPS treatment on GD15-16 or GD18-19 elicited deficits in reward prediction similar to those observed in schizophrenia. These results in conjunction with previous maternal immune activation research, suggest that timing of maternal infection is critical for the long-term outcome. Identifying which brain regions are critically altered during this period of neurodevelopment and how they are linked to negative symptoms warrants further investigation.

Schizophrenia is a debilitating mental disorder, which is diagnosed in 0.5-1.0% of the population (van Os, Kenis, & Rutten, 2010). The Diagnostic Statistic Manual of Mental Disorders (DSM5; American Psychiatric Association, 2013) defines schizophrenia as abnormalities in two or more of five domains: delusions, hallucinations, disorganised thinking, disorganised or abnormal motor behaviour, and negative symptoms. The prognosis for schizophrenia is frequently poor, with attempts to find preventive and curative treatments being largely unsuccessful (Mäkinen, Miettunen, Isohanni, & Koponen, 2008). The current primary treatment of schizophrenia, since their discovery in the 1950s, is antipsychotics. Antipsychotics have been proven effective in treating the positive symptoms of schizophrenia, although they have little influence on negative symptoms (Ellenbroek & Cesura, 2015). Through the use of animal models, researchers can hope to gain a better understanding of the negative symptoms of schizophrenia and treat them with greater efficacy.

Schizophrenia is a chronic disease with approximately 10% of sufferers eventually committing suicide (Kulhara, Banerjee, & Dutt, 2008). It is associated with enduring social, health, and financial burdens for the individual and also for families, caregivers, and society (Knapp, Mangalore, & Simon, 2004). Direct costs of the illness involve hospital and home nursing care, physician and health professionals, and medication. Indirect costs of schizophrenia involve lost productivity, with rates of employment amongst sufferers being 12.9% in France, 11.5% in the United Kingdom, and 30.3% in Germany (Gustavsson et al., 2011). Recently it has been calculated that the total annual cost of schizophrenia in Europe exceeds 93 billion Euro, coming second only to mood disorders (Gustavsson et al., 2011). Through greater understanding of the underlying mechanisms of schizophrenia, researchers can aid in the effective treatment of this disorder, reducing the societal costs and aiding the individual.

Receiving the diagnosis of schizophrenia generally peaks between the age of 10 and 25 for men and between 25 and 35 for women (Rajji, Ismail, & Mulsant, 2009). Onset of schizophrenia symptoms are typically preceded by a prodromal phase which heralds the approaching onset of the illness. The prodromal phase includes attenuated positive symptoms, such as illusions, ideas of reference, magical thinking; mood symptoms such as anxiety, dysphoria, and irritability; cognitive symptoms such as distractibility; and social withdrawal (Lieberman et al., 2001). As many of these symptoms overlap with a range of experiences and behaviours typical for the normal age group, they cannot be considered diagnostic. The development of unmistakable psychotic symptoms mark the formal onset of schizophrenia known as the active phase. This includes positive and negative symptoms, cognitive impairment, and social deficits. An average of one year will pass between psychotic onset and formal diagnosis and treatment, and an average of three years from the onset of prodromal symptoms (Lieberman et al., 2001) allowing the illness to progress prior to onset of treatment. Psychotic relapses, treatment failure, and incomplete recovery leads many of these individuals to follow a chronic course of the illness, experiencing persistent symptoms and decrements in functioning in comparison to their premorbid self (Lieberman et al., 2001). Research indicates that age and progression of schizophrenia is associated with a decrease in positive symptoms, while older individuals with schizophrenia tend to have persistent negative symptoms (Schultz et al., 1997).

Approximately 50-90% of individuals with schizophrenia will experience negative symptoms at any point in their illness, while 20-40% of these individuals will experience them permanently (Mäkinen et al., 2008). Negative symptoms are typically understood as features such as thoughts, behaviours, and emotions that are normally present but are reduced or absent as a result of the schizophrenia disease process (Ellenbroek & Cools, 2000; Mäkinen et al., 2008).

The negative symptom cluster includes *avolition* (diminished motivation and goal-directed behaviour), *anhedonia* (the reduced ability to experience pleasure), *asociality* (the reduced engagement in social activity and formation of relationships), *alogia* (poverty of spontaneous speech or content of speech), and *restricted affect* (blunted emotional expression) (Mäkinen et al., 2008).

Recent conceptualisations have placed these five core components of negative symptoms into two separate domains; diminished expressivity and diminished motivation (Strauss, Waltz, & Gold, 2014). The diminished expressivity dimension consists of restricted affect and alogia. The motivational dimension consists of avolition, anhedonia and asociality. The latter dimension incorporates deficits in reinforcement learning (Strauss et al., 2014), with an important distinction between consummatory and anticipatory pleasure (Gard, Kring, Gard, Horan, & Green, 2007). Consummatory pleasure refers to pleasure which is experienced when an individual is directly involved in an enjoyable activity, while anticipatory pleasure involves the recalling of previously pleasurable events and the predicting of future enjoyable events (Barnes, Der-Avakian, & Markou, 2014; Gard et al., 2007). Consummatory pleasure has been found to be normal in patients with schizophrenia, while deficits in anticipatory pleasure have been observed (Gard et al., 2007). Negative symptoms are considered a core aspect of schizophrenia phenomenology, with the motivational dimension theorised to be of greater importance to the core aspects of the illness and playing an integral role in life-course trajectories, including subjective quality of life, functional and occupational outcomes, and recovery (Strauss et al., 2014). Reductions in pleasure seeking and motivational capabilities have not only been suggested as the most disabling facets of schizophrenia, but are theorised to underpin other negative symptoms including social withdrawal and cognitive impairments (Barnes et al., 2014).

To develop effective therapeutics, especially in regards to negative symptoms, it is crucial to understand the underlying mechanisms and associated risk factors leading to the development of this debilitating disorder.

Little is known regarding the aetiology of schizophrenia, although family studies show a clear heritable genetic component with the concordance rate for monozygotic twins reaching 50% (Ellenbroek & Yoon, 2016). The exact nature of this component is unknown, no single gene has been identified that can explain a significant portion of the variance in the illness, leading to the polygenic theory of schizophrenia. As outlined by the concordance rate, the development of schizophrenia is not entirely due to genetic transmission, the environment plays a crucial role indicating a gene-environment interaction: genetic factors operate through making individuals vulnerable to environmental risks (van Os et al., 2010).

Developmental risk factors have been identified in the later development of schizophrenia, including growing up in an urbanised area, being part of a minority group, illicit drug use, and developmental trauma (van Os et al., 2010). Minority group effects are evident in both first and second generation migrants, with a reduced risk when there is an increased proportion of the individual's own ethnic group living in the occupied area (van Os et al., 2010). The risk associated with growing up in an urbanised area may incorporate both minority effects and illicit cannabis use as cities have higher populations and higher rates of drug use (van Os et al., 2010), although social aspects including trust, bonding, and safety in neighbours are proposed as having a developmental impact on the mental health of children in urban contexts (Krabbendam & Van Os, 2005). Such developmental risk factors are not entirely understood due to the high number of dynamic environmental factors involved. Risk factors leading to the later

development of schizophrenia have also been identified in earlier developmental stages, including prenatal environmental exposures.

The neurodevelopment theory of schizophrenia incorporates prenatal environmental risks, proposing that disturbances in early neurodevelopment are a possible causal factor in the disorder (Khandaker, Zimbron, Lewis, & Jones, 2013). Prenatal environmental risks fitting well with this theory include maternal stress, infection, nutritional deficiency, low and high neonatal vitamin D, toxoplasmosis, and various other pregnancy and birth complications (van Os et al., 2010). Exposure to infectious agents during pregnancy have been identified as a risk factor in adult offspring (Brown & Derkits, 2010), this is a factor of great interest due to the high number of infectious agents universally present. Research into this area began with a report investigating the higher rates of schizophrenia found in individuals thought to be exposed prenatally to the Helsinki influenza epidemic of 1957 (Mednick, Machon, Huttunen, & Bonett, 1988). The results of this investigation revealed that individuals exposed to the viral epidemic during the second trimester of gestation had an increased risk of being admitted into psychiatric care with a diagnosis of schizophrenia (Mednick et al., 1988). A limiting aspect of this study is that it used a population-based design, and therefore it lacked conformation of a maternal infection for each individual case.

Since interest into this area was sparked, a wide variety of infections have been identified in connection with later development of schizophrenia, including influenza, herpes, polio, respiratory infections, and rubella (Clarke, Tanskanen, Huttunen, Whittaker, & Cannon, 2009). Such a variety indicates a pathogenic mechanism which is common to many infections (Clarke et al., 2009). Serological confirmation of prenatal exposure to infection have been examined in relation to mechanisms associated in the development of schizophrenia and other psychoses

(Buka, Tsuang, Torrey, Klebanoff, Bernstein, et al., 2001). Using a cohort of the Collaborative Perinatal Project, stored blood samples taken from mothers at the end of pregnancy were retrieved. These were analysed for immunoglobulins and antibodies which were then found to be related to increased risk of schizophrenia development. Cytokine levels, particularly tumour necrosis factor alpha (TNF- α) have also been implicated using the Collaborative Perinatal Project (Buka, Tsuang, Torrey, Klebanoff, Wagner, et al., 2001). Cytokines are a category of cell signalling proteins which are involved in inflammation. TNF- α is a critical cytokine for the onset and magnification of the inflammatory process in response to infectious agents or injury (Boin et al., 2001). This led the authors to hypothesise that increased levels of cytokines exert a direct effect on the developing brain, potentially leading directly to schizophrenia development, or later interacting with environmental factors which lead to its development (Buka, Tsuang, Torrey, Klebanoff, Wagner, et al., 2001).

Cytokines are involved in normal central nervous system function, acting directly on neural cells or modulating neurotransmitter systems (Boin et al., 2001). TNF- α can exert both neurotrophic and neurotoxic effects, influencing neural cell growth and proliferation, with polymorphisms in TNF- α genes also being associated with schizophrenia (Boin et al., 2001; Brown & Derkits, 2010). Findings such as these indicate that infection induced elevation of cytokine levels can interfere with neurodevelopmental events during foetal life (Meyer et al., 2006) although timing of infection has yielded inconsistent results. Evidence has identified the second trimester as a critical period for viral infections and influenza while the first trimester was related to influenza and especially rubella infection (Boksa, 2010). Human studies have provided important evidence that infections can alter neurodevelopment and increase the risk of schizophrenia. However, research utilising human participants lacks the capacity to infer causal

relationships (Meyer et al., 2006), therefore, the adoption of animal models of schizophrenia is imperative in determining these relationships.

It has been estimated that 14-21% of schizophrenia cases are directly caused by maternal infection, with the risk associated with infection being threefold to sevenfold more in genetically susceptible individuals (Boin et al., 2001). Developing an animal model based on this evidence offers a significant opportunity to provide a better understanding for disease aetiology, and the development and testing of effective treatment strategies, especially in regards to poorly treated negative symptoms (Missault et al., 2014). The maternal immune activation (MIA) model induces immune activation in pregnant animals, mainly rodents, allowing for the assessment of its role played on the development of schizophrenic like outcomes in offspring (Wolff, Cheyne, & Bilkey, 2011). Preclinical studies using this model have reported neurophysiological, neurochemical, and behavioural alterations related to those found in individuals diagnosed with schizophrenia (for review see Boksa, 2010; Brown & Derkits, 2010). Two agents are commonly used to induce an immune response, polyinosinic:polycytidylic acid (polyI:C) a synthetic analogue of double-stranded RNA used to mimic viral infection, and lipopolysaccharide (LPS) a bacterial wall endotoxin used to mimic bacterial infection (Brown & Derkits, 2010). These have been injected using varying doses intraperitoneally, intravenously, or subcutaneously, and range from a single injection to daily injections throughout the gestational period (Boksa, 2010). Although studies vary significantly in the type, dose, and timing of immunogen administration, results clearly indicate that exposure to immune activation causes acute and long-lasting changes in behaviour and central nervous system function and structure related to schizophrenia (for review Boksa, 2010; Brown & Derkits, 2010).

The large diversity in protocols, however, implies that the MIA model requires further investigation to select the best strategy to induce MIA. A rat's gestational period is approximately 22 days, and studies using LPS or polyI:C have generally focussed on the second half of pregnancy. Fortier, Luheshi, and Boksa (2007), for instance, used gestational days (GD) 10-11, 15-16, and 18-19, representing early, middle, and late stages of rat pregnancy which coincide with different neurodevelopmental events, particularly in regard to the dopaminergic system. Fortier et al. (2007) investigated the effects of prenatal exposure to LPS, polyI:C, and turpentine administered during these three periods of gestation. The authors found that exposure of LPS and turpentine (but not polyI:C) on GD15-16 led to deficits in prepulse inhibition of the acoustic startle response, while exposure on GD10-11 led to high rates of offspring and dam death. In search of the optimal paradigm for inducing cognitive deficits, a recent study also compared these three timing protocols using subcutaneously injected 0.5mg/kg LPS to model maternal infectious agent exposure (Waterhouse, Roper, Brennan, & Ellenbroek, 2016). It was found that animals exposed to maternal LPS treatment on GD10-11 showed a significant deficit in prepulse inhibition, latent inhibition, and T-maze learning, relating to the deficits in sensorimotor gating, attention, and working memory seen in patients with schizophrenia. These deficits did not occur on GD15-16 or GD18-19, suggesting that this early period of gestation is sensitive to maternal LPS exposure specifically within cognitive domains. However, the same study did find some cognitive deficits on GD18-19 (notably a deficit in conditioned taste aversion), suggesting different long-term alterations depending on the gestational exposure to LPS.

Both cognitive and negative symptoms are evident during the early stages of schizophrenia development, neither are substantially improved with antipsychotic medication,

and both have a high impact on functional outcomes (Foussias & Remington, 2010; Harvey, Koren, Reichenberg, & Bowie, 2006). While cross-sectional evidence has found negative symptoms correlate with several domains of cognitive functioning, a longitudinal investigation by Bell and Mishara (2006) failed to establish a relationship between progression of negative symptoms and neurocognitive function, concluding that they likely represent independent disease processes. Pathological changes in separate brain regions have been proposed in negative and cognitive symptoms (Harvey et al., 2006). Dysfunction in the dorsolateral prefrontal cortex has been related to cognitive symptoms (Perlstein, Carter, Noll, & Cohen, 2001) while hypoactivity within the anterior cingulate cortex has been associated with negative symptomology (for review see Bersani et al., 2014). In regards to the MIA model, these differing structures related to cognitive and negative symptoms may lead to distinct periods of sensitivity during neurodevelopment.

Current Study

Aims and Hypothesis

The current study aims to further validate the MIA model, focussing specifically on the negative symptoms. As previously mentioned, the negative symptoms of schizophrenia are not affected by the currently prescribed antipsychotic medication (Ellenbroek & Cesura, 2015). Identifying behavioural abnormalities related to the motivational dimension of negative symptoms, which play a crucial role in the progression and outcome of schizophrenia, would provide further face validity for the MIA model, allowing for the development and testing of novel efficacious treatment strategies (Barnes et al., 2014). Anticipatory pleasure is operationalised through the behavioural assays anticipatory locomotor activity and successive contrast. The anticipatory locomotion experiment is used to assess reward prediction. This

experiment involves the measuring of locomotor activity prior to the conditioned presentation of a reward, with studies consistently demonstrating that after repeated presentation of a reward animals exert increased locomotor activity. Rats show this effect with increased rearing prior to the presentation of palatable food (Mendoza, Angeles-Castellanos, & Escobar, 2005), and male rats demonstrate anticipatory exploring behaviour prior to the conditioned introduction of a female (Pfaus & Phillips, 1991). Significantly reduced anticipatory locomotor activity is believed to reflect an impairment in reward prediction (Barnes et al., 2014). The successive contrast experiment is also used to assess reward prediction, through the behaviour elicited in response to an alteration in a highly desirable and anticipated reward (Barnes et al., 2014). This experiment involves measuring consummatory behaviour of animals shifted from a high value reward (e.g. 32% sucrose solution) to a low value reward (e.g. 4% sucrose), animals should consume less of the low value reward in comparison to subjects that have only experienced this low value reward (Barnes et al., 2014). This phenomena is termed successive negative contrast (SNC; Pellegrini & Mustaca, 2000). A non-significant SNC effect is believed to reflect an impairment in reward prediction.

The current study will follow the same LPS administration protocol as Waterhouse et al. (2016) and conduct anticipatory locomotion and successive contrast experimental measures of anticipatory pleasure in offspring. It is hypothesised that offspring exposed to maternal immune response on one of the three gestational periods will elicit impaired anticipatory locomotor activity and reduced SNC effects in comparison to those exposed to saline during foetal development. Due to the differing structures and processes involved in the negative and cognitive domains of schizophrenia, the sensitive periods of gestation for each of these domains may differ and not coincide with the results found by Waterhouse et al. (2016). Identifying a

period of gestation sensitive to maternal immune exposure specific to the negative symptom domain of schizophrenia will further validate the MIA model.

Method

Subjects

Test subjects were male Sprague-Dawley rats that were either exposed to the bacterial endotoxin LPS (n=47, 8 litters) or to prenatal saline (n=26, 5 litters). All dams and their offspring were bred in the vivarium at Victoria University of Wellington. Animals were weaned at 21 days old, separated by sex, and housed in groups of 2 to 5 animals in standard polycarbonate cages. Water was available *ad libitum*. The housing room had a controlled temperature of 19-21°C and 55-60% humidity. The housing facility was on a reversed 12-hour light dark cycle with lights on at 1900hrs. Radio-frequency identification transponders were implanted subcutaneously between post-natal day (PND) 40 and 70, allowing for accurate animal identification. These were obtained from Kent Scientific Corporation in the United States (product ID-100US). Behavioural testing took place after animals reached adulthood at PND60. Animals were handled and tested on social measures by another experimenter between PND21-25, 30-35, 60-65, and 70-80. All procedures were approved by the Victoria University of Wellington Animal Ethics committee.

All animals were placed on food deprivation five days prior to the onset of the experiment, and this schedule was maintained throughout the experimental procedures. During this time food intake was limited to 15g per day, reducing body weight to approximately 85% of their free feeding weight.

Drug Treatments

Prenatal treatment. LPS (from *Escherichia coli* 0111:B4 purified by phenol extraction, Sigma-Aldrich, Auckland New Zealand) was diluted in saline (0.5mg/ml). Pregnancy was established through mating rats overnight, the presence of a vaginal plug marked that day as GD0. Pregnant female rats were injected subcutaneously once daily on two consecutive days with either LPS (0.5mg/kg) or saline. Injections took place at one of the following gestational periods: GD10-11, 15-16, or 18-19. For group size information refer to Table 1.

Table 1.

Group sizes per treatment group

<u>Gestation Day (GD)</u>	<u>Treatment Group</u>	<u>Group Size (n)</u>
10-11	LPS1	17
	SAL1	4
15-16	LPS2	18
	SAL2	11
18-19	LPS3	12
	SAL3	11

Anticipatory Locomotion Experiment

Apparatus. The anticipatory locomotion experiment was conducted in eight locomotor activity chambers (Med Association Inc., USA; model ENV-515) made from plexiglass (dimensions: 42 x 42 x 30 cm) set in sound attenuated boxes. Each of the internal walls of the chambers were equipped with two banks of eight photoelectric infrared sensors. Each of the sensors were 2.5cm apart and 2.5cm off the chamber floor. All chambers were connected to a computer which recorded activity through the following variables: rearing (breaking high placed beams), ambulatory counts (cross-over between the lower beams), and total activity (total horizontal beam breaks for sensor).

During testing a red light was illuminated outside the boxes and white noise was continuously present to mask any outside disturbances. After each test session the animals were returned to their home cages and the chambers were cleaned and wiped with 70% ethanol.

Procedure. The experiment included two phases. The first phase was the habituation phase; animals were put into the locomotor activity chambers for 60 minutes daily for five days. Activity data was recorded at five minute intervals for the first 30 minutes. During this phase animals were fed at random intervals (2-6 hours) after testing. The second phase was the anticipation phase; animals were again placed in the chambers for 60 minutes daily for five days. Activity data was recorded at five minute intervals during the first 30 minutes. In this phase the animals were fed in their chambers 30 minutes after being placed in the chamber.

Successive Negative Contrast

Apparatus. The SNC experiment was conducted in the animals' home cages. 32% sucrose solution was prepared by mixing 32 grams of white sugar for every 68 grams of distilled water. 4% sucrose solution was prepared by mixing 4 grams of white sugar for every 96 grams of distilled water. The solutions were administered using white plastic drinking bottles with non-leak stoppers placed in the drinking tubes. The non-leak stoppers were used to prevent dripping to ensure intake calculations were as accurate as possible.

Procedure. Following directly from the former experiment, animals were housed separately in new home cages and randomly assigned to either the 32-4 group or the 4-4 group. Animals in the 32-4 group received the 32% sucrose solution during the pre-shift phase and the 4% solution during the post-shift phase. Animals in the 4-4 group received the 4% solution during the pre-shift phase and the 4% solution again in the post-shift phase. Two days prior to

the onset of the experiment, all animals were given access to their assigned solutions in their home cages for one hour.

The experiment included two phases; the pre-shift and post-shift phase. During the pre-shift phase animals were given access to their assigned solution for five minutes per day for ten days. During the post-shift phase both groups of animals were given access to 4% sucrose solution for five minutes per day for eight days.

Statistical Analyses

Data analysis was performed using a mixed measures analysis of variance (ANOVA). Statistical analyses used for each experiment will be explained in detail in the results section.

Results

Due to there being no main effect between the saline conditions ($p = .613$), these were grouped into one overall saline condition (SAL).

Anticipatory locomotor activity

Correlational analyses were run between ambulatory counts and distance travelled to assess the justification of including both measures. As the analysis revealed the variables were highly related ($r = .998$, $p < .001$), ambulatory counts were removed before further analysis was undertaken.

Statistical analysis. An initial 4 (treatment: LPS1 vs. LPS2 vs. LPS3 vs. SAL) x 5 (habituation day: 1-5) mixed-measures ANOVA was run with treatment as a between-subjects factor and habituation day as a repeated-measure. A similar 4 (treatment: LPS1 vs. LPS2 vs. LPS3 vs. SAL) x 5 (anticipation day: 1-5) mixed-measures ANOVA was run with treatment as a

between subjects factor and anticipation day as a repeated-measure. Significant interactions found in these analyses were followed up using a 2 (treatment: LPS1, LPS2, or LPS3 vs. SAL) x 5 (habituation day: 1-5) mixed measures ANOVA or a 2 (treatment: LPS1, LPS2, or LPS3 vs. SAL) x 5 (anticipation day: 1-5) mixed measures ANOVA. When Mauchly's Test for Sphericity indicated that the assumption of sphericity had been violated, a Greenhouse-Geisser correction was applied.

Distance travelled. During the habituation phase, main effects were found for day ($F(3.28, 226.29) = 71.191, p < .001$) and treatment ($F(3, 69) = 2.923, p = .04$). A significant interaction was found between treatment and habituation day ($F(9.84, 226.29) = 1.994, p = .025$). Further analyses showed LPS3-SAL significantly interacted with habituation day ($F(3.22, 115.85) = 4.057, p = .007$), refer to figure 1. Follow-up analysis show significant differences for habituation day 2 ($t(36) = 3.43, p = .001$), day 3 ($t(36) = 4.45, p = .001$), and day 4 ($t(36) = 2.35, p = .024$), but not days 1 and 5 ($p > .05$). This suggests that LPS3 travelled significantly more distance than saline on day 2 ($M_{LPS3} = 2022.34$ vs $M_{SAL} = 1328.83$), on day 3 ($M_{LPS3} = 1983.27$ vs $M_{SAL} = 1249.51$), and on day 4 ($M_{LPS3} = 1704.09$ vs $M_{SAL} = 1252.31$).

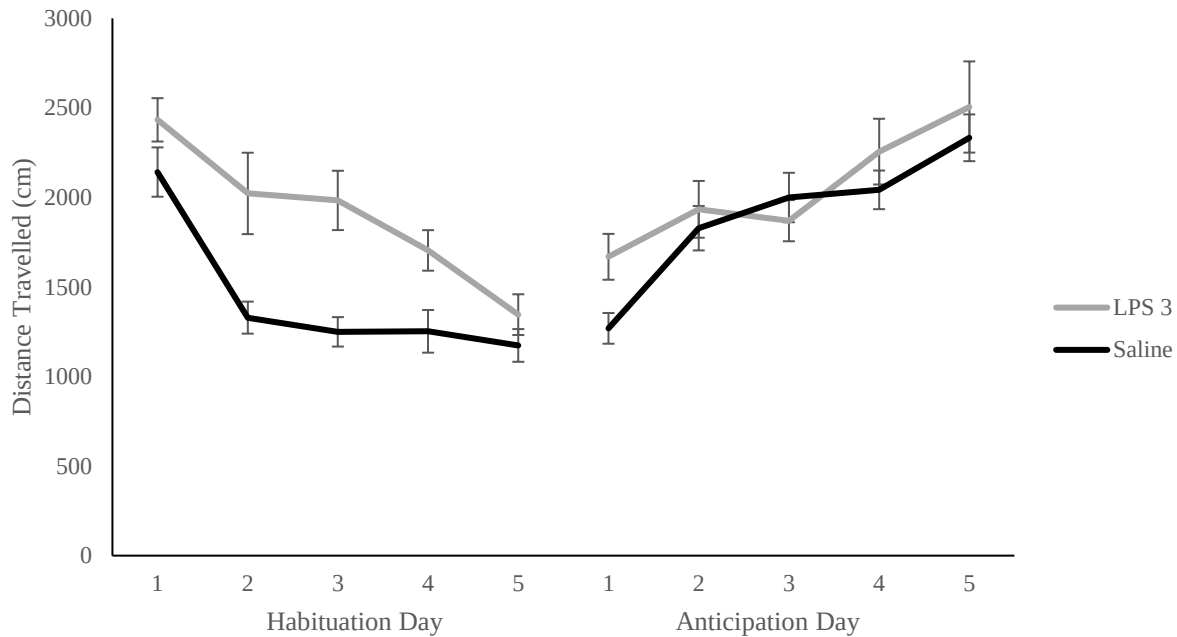


Figure 1. Distance travelled by LPS3 and Saline conditions during the habituation and anticipation phases. Vertical lines represent standard error of the mean.

During the anticipation phase, a main effect was found for day ($F(3.39, 234.16) = 44.452, p < .001$) and no main effect was found for treatment ($p = .355$). A significant interaction was found between treatment and anticipation day ($F(10.18, 234.16) = 2.027, p = .031$). Follow-up analyses showed that LPS2-SAL had a close to significant interaction with anticipation day ($F(2.93, 122.89) = 2.231, p = .089$), refer to figure 2.

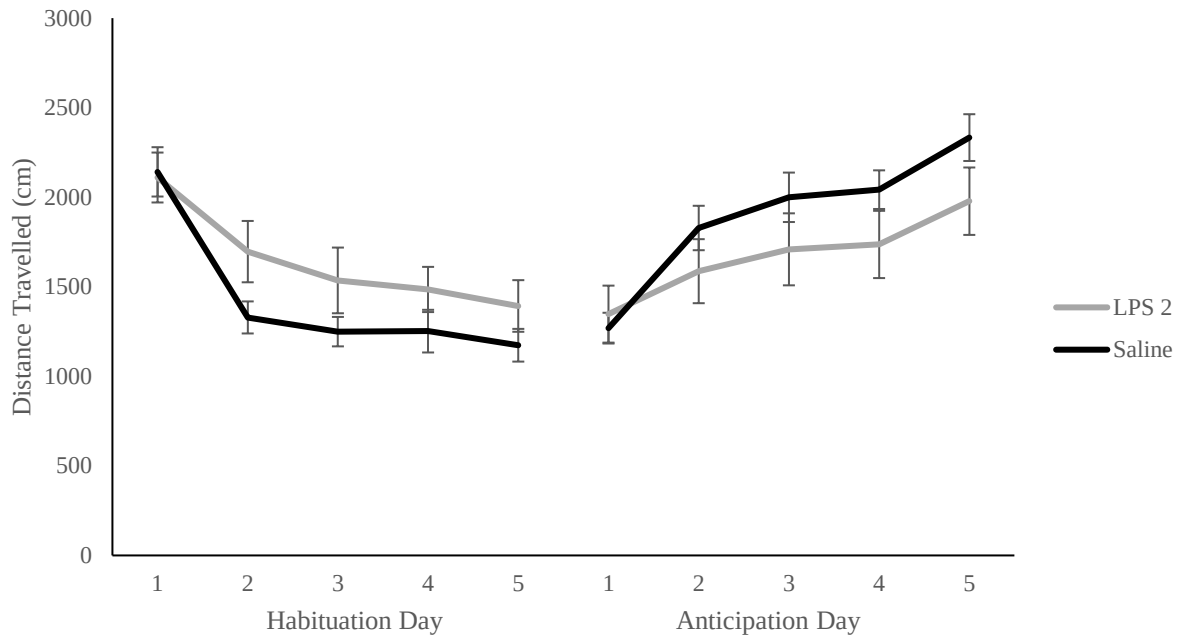


Figure 2. Distance travelled by LPS2 and Saline conditions during the habituation and anticipation phases. Vertical lines represent standard error of the mean.

Rearing. During the habituation phase, main effects were found for day ($F(3.02, 208.4) = 100.380, p < .001$) and treatment ($F(3, 69) = 5.879, p < .001$). A significant interaction was found between treatment and habituation day ($F(9.06, 208.4) = 3.209, p < .001$). Further analyses showed that only LPS3-SAL significantly interacted with habituation day ($F(1, 36) = 11.097, p = .002$) refer to figure 3. Follow-up analysis show significant differences for habituation day 1 ($t(36) = 3.502, p = .001$), day 2 ($t(36) = 3.568, p = .001$), day 3 ($t(36) = 2.317, p = .026$), and day 4 ($t(36) = 2.079, p = .045$), but not day 5 ($p = .635$). This suggests that LPS3 reared significantly more than saline on day 1 ($M_{LPS3} = 186.83$ vs $M_{SAL} = 130.96$), day 2 ($M_{LPS3} = 129.83$ vs $M_{SAL} = 76.69$), on day 3 ($M_{LPS3} = 123.17$ vs $M_{SAL} = 90.97$), and on day 4 ($M_{LPS3} = 95.67$ vs $M_{SAL} = 68.69$).

During the anticipation phase, main effects were found for day ($F(3.2, 220.52) = 37.247$, $p < .001$) and treatment ($F(3, 69) = 3.032$, $p = .035$). No significant interaction was found between treatment and anticipation day on rearing count ($p = .838$).

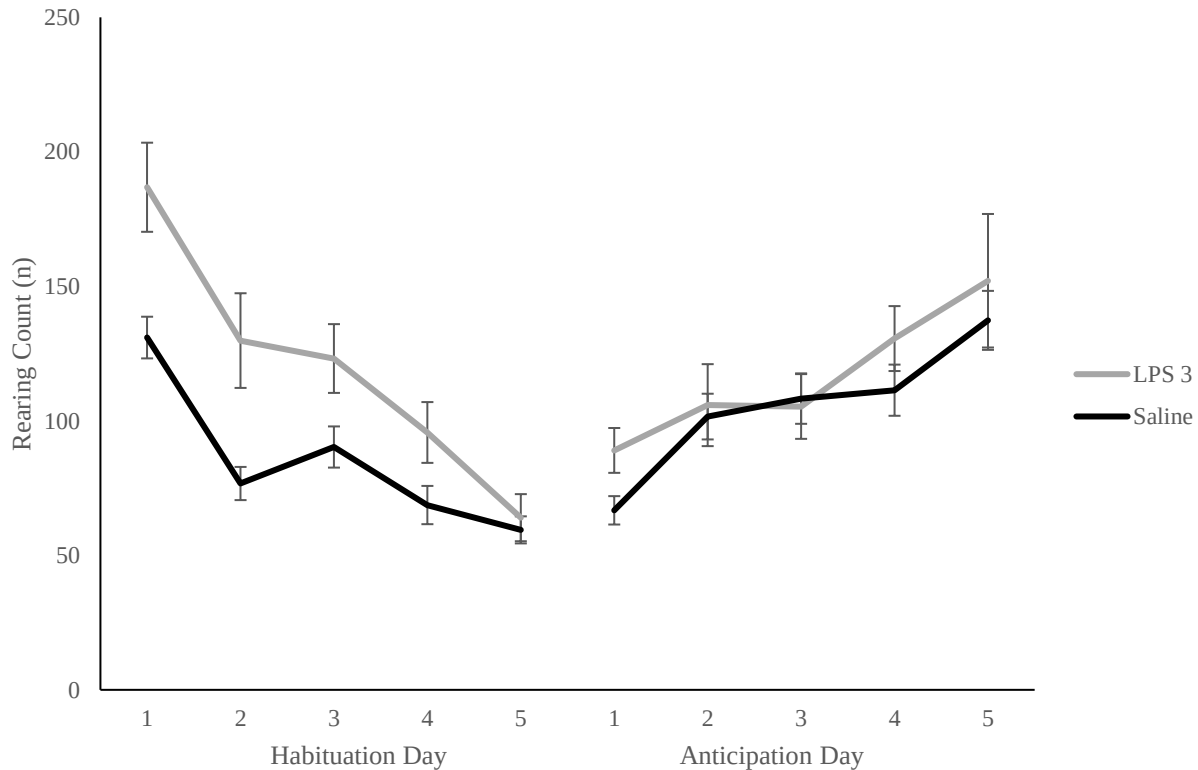


Figure 3. Rearing counts of LPS3 and saline during the habituation and anticipation phases. Vertical lines represent standard error of the mean.

Successive Negative Contrast

Statistical analyses. Results were analysed within treatment conditions. An initial 2 (solution: 4-4 vs. 32-4) $\times$ 10 (pre-shift day: 1-10) mixed-measures ANOVA was run with solution as a between-subjects factor and pre-shift day as a repeated-measure on daily consumption. A similar 2 (solution: 4-4 vs. 32-4) $\times$ 10 (post-shift day: 1-8) mixed-measures ANOVA was run with solution as a between subjects factor and post-shift day as a repeated-

measure on daily consumption. These were performed to show whether 32-4 group consumed greater or equal quantities of solution compared to the 4-4 group within their treatment conditions. A one-way ANOVA with solution (4-4 vs. 32-4) as a between subject factor assessed on post-shift day 1 for a negative contrast. These were identified by significantly less consumption of the 4% solution by the 32-4 group in comparison to their 4-4 controls.

Saline treatment. Refer to figure 4. Statistical analysis of the saline treatment revealed no significant interaction ($p = .766$) or main effect of solution ($p = .301$), though a main effect was found for day ($F(5.12, 216) = 8.975, p < .001$). The latter finding was not subject to post-hoc analysis as it suggests the 32-4 and 4-4 had similar intake in response to pre-shift phase day. Post-shift analysis revealed a significant main effect of day ($F(3.89, 168) = 17.327, p < .001$), interaction between solution and day ($F(3.89, 168) = 7.295, p < .001$), and no main effect of solution ($p = .294$).

Solution groups were significantly different on post-shift day 1 ($F(1, 24) = 30.005, p < .001$), such that the 32-4 group ($M = 5.21, SD = 2.36$) consumed less than the 4-4 group ($M = 9.42, SD = 1.31$), demonstrating a SNC effect.

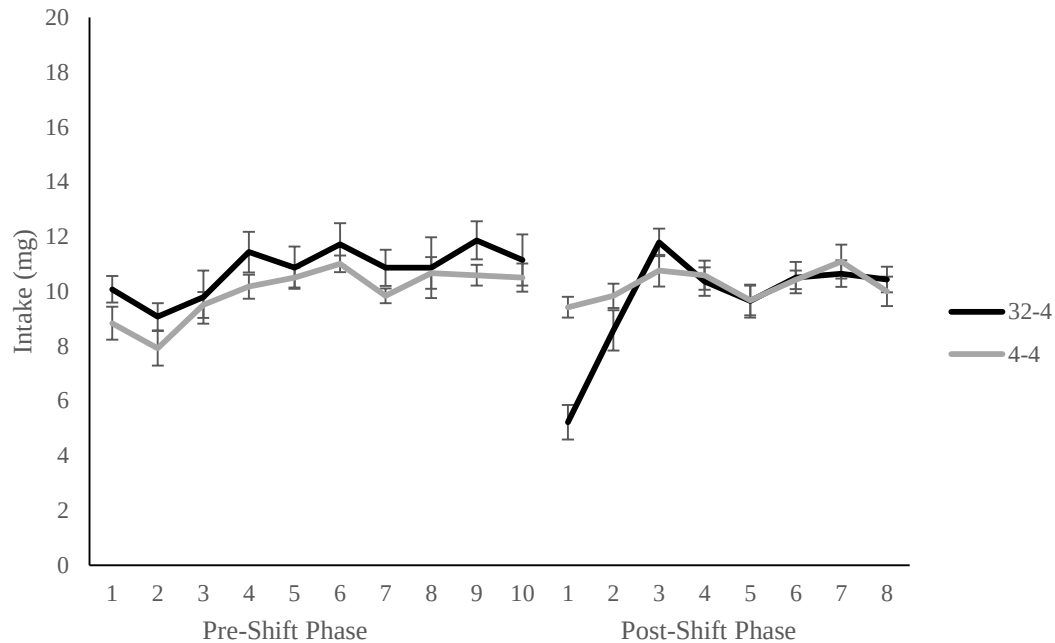


Figure 4. Sucrose intake in the prenatally treated saline condition. Vertical lines represent standard error of the mean

LPS1 treatment. Refer to figure 5. Statistical analysis of the LPS1 treatment during the pre-shift phase revealed a significant main effect of day ($F(9, 135) = 8.63, p < .001$), and a main effect of solution ($F(1, 15) = 22.848, p < .001$), such that the 32-4 group ($M = 11.84, SD = 1.38$) consumed more than the 4-4 group ($M = 9.14, SD = 1.1$). No interaction was found between the two factors ($p = .224$). Analysis of the post-shift phase revealed a significant main effect of day ($F(3.56, 105) = 11.826, p < .001$) and solution ($F(1, 15) = 4.558, p = .05$), and a significant interaction ($F(3.56, 105) = 10.07, p < .001$).

Solution groups were significantly different on post-shift day 1 ($F(1, 15) = 6.731, p = .02$), such that the 32-4 group ($M = 5.3, SD = 2.91$) consumed less than the 4-4 group ($M = 8.43, SD = 1.51$), demonstrating a SNC effect.

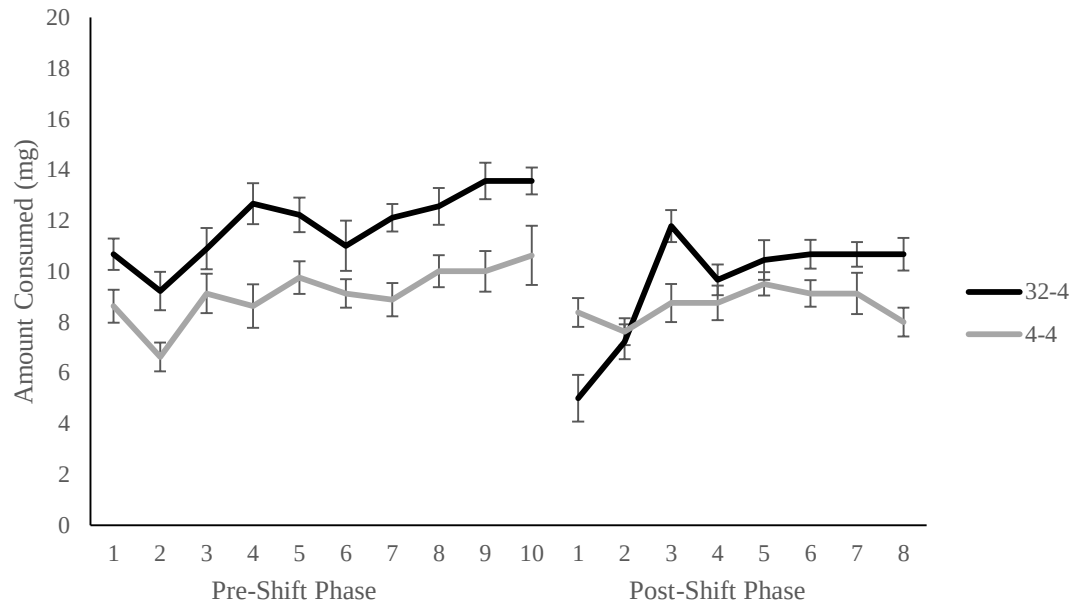


Figure 5. Sucrose intake in the prenatally treated LPS1 condition. Vertical lines represent standard error of the mean.

LPS2 treatment. Refer to figure 6. Statistical analysis of the LPS2 treatment during the pre-shift phase revealed a significant main effect of day ($F(3.71, 135) = 18.34, p < .001$), and a main effect of solution ($F(1, 15) = 39.652, p < .001$), such that the 32-4 group ($M = 14.89, SD = 1.98$) consumed more than the 4-4 group ($M = 9.36, SD = 2.08$). No interaction was found between the two factors ($p = .088$). Analysis of the post-shift phase revealed a significant main effect of day ($F(2.99, 105) = 4.413, p = .008$), and a main effect of solution ($F(1, 15) = 4.579, p = .049$). No interaction was found between the two factors ($p = .224$).

Solution groups were not significantly different on post-shift day 1 ($p = .62$), failing to demonstrate a SNC.

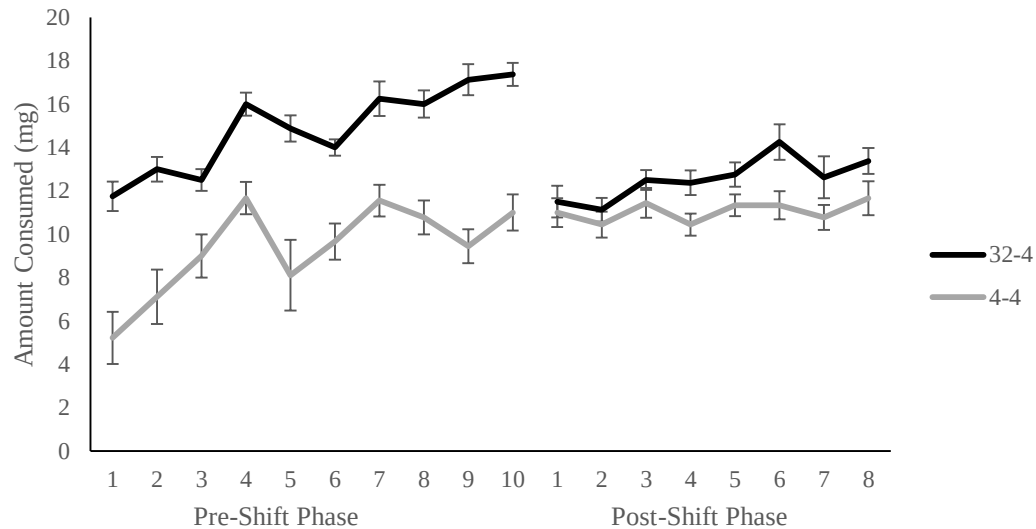


Figure 6. Sucrose intake in the prenatally treated LPS2 condition. Vertical lines represent standard error of the mean.

LPS3 treatment. Refer to figure 7. Statistical analysis of the LPS3 treatment during the pre-shift phase revealed a significant main effect of day ($F(2.55, 90) = 13.33, p < .001$), and a main effect of solution ($F(1, 10) = 92.574, p < .001$), such that the 32-4 group ($M = 14.2, SD = 2.41$) consumed more than the 4-4 group ($M = 9.05, SD = 1.63$). A significant interaction occurred between the two factors ($F(2.55, 90) = 3.382, p = .04$). Analysis of the post-shift phase revealed a significant main effect of day ($F(7, 70) = 9.136, p < .001$), and no main effect of solution ($p = .085$). A significant interaction between the two factors occurred ($F(7, 70) = 6.582, p < .001$).

Solution groups were not significantly different on post-shift day 1 ($p = .687$), failing to demonstrate a SNC.

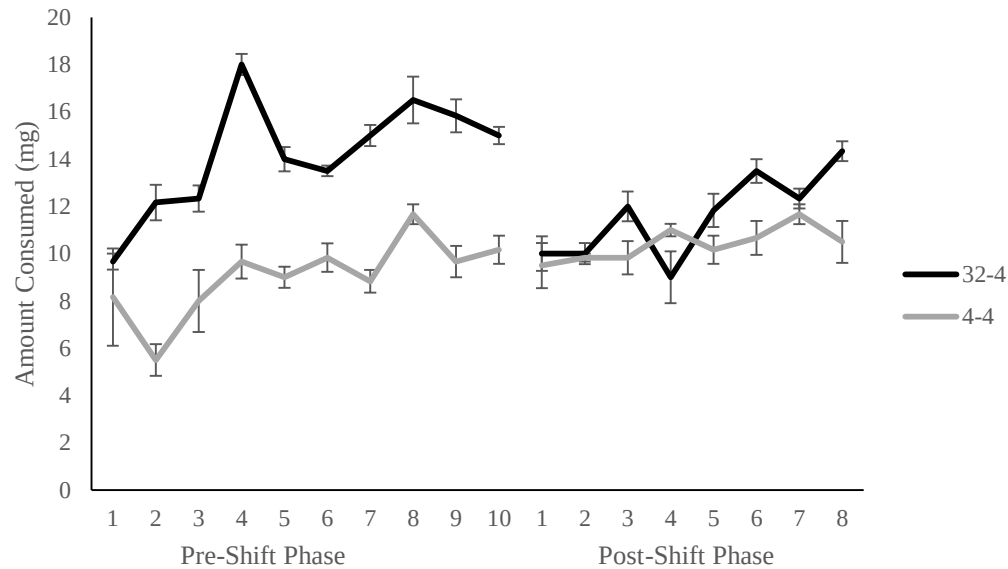


Figure 7. Sucrose intake in the prenatally treated LPS3 condition. Vertical lines represent standard error of the mean.

Discussion

Negative symptoms are considered a core aspect of schizophrenia. Current pharmacological treatment options show little efficacy in the treatment of the negative symptomatology of schizophrenia (Ellenbroek & Cesura, 2015). Negative symptoms can be separated into two domains; diminished motivation and diminished expressivity (Strauss et al., 2014). The diminished motivation domain is theorised to play an integral role in life-course trajectories, including impaired functional outcomes. Contributing to this domain are deficits in anticipatory pleasure, which are reflected by disruptions in reward perception and anticipation, which is manifested as a reduced engagement in reward-seeking behaviour (Gard et al., 2007; Vanden Broeke, 2016). Exposure to maternal infection is a neurodevelopmental risk factor in the later onset of schizophrenia, with 14-21% of cases thought to be directly caused by this exposure (Boin et al., 2001). The MIA model reproduces this risk factor through inducing immune activation in pregnant animals, allowing for the assessment of its role in the development of

schizophrenic-like outcomes in offspring (Wolff et al., 2011). However, large diversity in protocols used imply that the MIA model requires further investigation to determine the best strategy to induce an immune response. The primary aim of this study was to further validate the MIA model of schizophrenia, with a focus on reward anticipation as a measure within the diminished motivation domain of negative symptoms.

Following the protocol used by Waterhouse et al. (2016), the current study induced MIA by injecting pregnant rats with the bacterial endotoxin LPS or saline at one of three gestational periods: GD10-11, GD15-16, or GD18-19. Reward anticipation was measured in adult offspring using the behavioural assay of anticipatory locomotion and SNC. It was hypothesised that offspring exposed to a MIA at one of these three periods would elicit reduced SNC effects and impaired anticipatory locomotor activity in comparison to the offspring exposed to saline.

Within the anticipatory locomotion paradigm it was found that when compared to controls, animals exposed to LPS treatment on GD18-19 show increases in both distance travelled and rearing measures. The higher rates of movement and rearing would suggest that there is no deficit in behavioural responding (i.e. motor impairments), but may suggest deficits in cognitive flexibility, which is consistent with habituation deficits observed in individuals diagnosed with schizophrenia (Geyer, Swerdlow, Mansbach, & Braff, 1990). Consistent with this finding, using specific measures of cognitive symptoms, Fortier et al. (2007) observed significant deficits in prepulse inhibition in animals exposed to MIA on GD15-16 and GD18-19.

The hypothesis that animals exposed to maternal infection would show reduced anticipatory behaviour was supported. During the anticipation phase of this paradigm, animals exposed to LPS treatment on GD15-16 travelled less in comparison to those exposed to saline. Although this difference was not statistically significant, there was a general tendency to elicit

decreased anticipatory behaviour prior to the presentation of the conditioned reward. While there is a lack of anticipatory locomotion research on the MIA model of schizophrenia, these findings are consistent with investigations on the role of dopamine on alterations in reward prediction. Vanden Broeke (2016) found that D₁ mutant rats, with a point mutation in the dopamine D₁ receptor leading to a 50% reduction in D₁ binding, have impaired anticipatory responses to an expected reward. Other research also support the role of dopamine in reward prediction, the administration of dopamine antagonists have been found to induce a reduction in anticipatory locomotor activity (Barbano & Cador, 2006). While the current study did not focus on the role of dopamine in schizophrenia, these findings may suggest a possible dopaminergic neurodevelopmental effect as the result of MIA exposure with then lead to a reduction in reward anticipation.

The SNC effect is also considered a measure of reward prediction (Barnes et al., 2014). In support of our hypothesis, it was found that animals exposed to LPS treatment on both GD15-16 and GD18-19 failed to demonstrate a SNC effect. These animals did not significantly reduce their intake following the downshift from 32% to 4% sucrose solution in comparison to those that had only received the 4% sucrose solution. Although there is a lack of SNC literature on the MIA model of schizophrenia, research into the role of dopamine on the SNC effect has yielded interesting results. Amphetamine (indirect dopamine agonist) administration has been shown to attenuate SNC effects (Phelps, Mitchell, Nutt, Marston, & Robinson, 2015), while amphetamine withdrawal has been shown to exaggerate SNC effects (Barr & Phillips, 2002).

Overall, results of the current study supported our hypothesis. Animals exposed to LPS treatment on GD15-16 or GD18-19 were found to show a reduction in reward anticipation comparable to those seen in schizophrenia. This study is largely consistent with other studies

using immune activating agents such as LPS, polyI:C, and turpentine (e.g. Boksa, 2010; Fortier et al., 2007; Waterhouse et al., 2016). Using the same LPS administration protocol, Waterhouse et al. (2016) investigated the cognitive symptom domain of schizophrenia. They found that animals exposed to LPS treatment on GD10-11 elicited deficits comparable to the cognitive symptomology of schizophrenia. The notion that the structures of the brain develop sequentially may provide an explanation as to why the timing of a maternal response is critical to the differing long-term outcomes relevant to schizophrenia (Workman, Charvet, Clancy, Darlington, & Finlay, 2013). Later periods of gestation may have a sensitivity in the development of reward anticipation deficits similar to those observed in the diminished motivation domain of negative symptoms.

The cytokine hypothesis contends that elevated cytokine levels, due to infection and other complications, interfere with structural and phenotypic development of the brain (Watanabe, Someya, & Nawa, 2010). This may offer an explanation to the pathological changes in different brain regions that have been associated with negative and cognitive symptomology of schizophrenia (Harvey et al., 2006). Dysfunction in the dorsolateral prefrontal cortex has been related to cognitive symptoms (Perlstein et al., 2001), while dysfunction in the anterior cingulate cortex has been related to negative symptomology (Bersani et al., 2014). The anterior cingulate cortex has been implicated in assessing whether an effort or action is worth taking for an expected benefit (Schweimer & Hauber, 2006). Human and animal research indicates that dopamine input and D₁ receptors within the anterior cingulate cortex are essential in this form of decision making (Schweimer & Hauber, 2006). Recent research within our lab has also provided support for the role of D₁ receptors in observable deficits in reward prediction (Vanden Broeke, 2016). A neurodevelopmental disturbance in the dopaminergic function within the anterior

cingulate cortex may offer an explanation of the deficits observed in the current study, thus offering a potential explanation into the function of MIA exposure.

Furthermore, although not directly relevant to the SNC effect, during the pre-shift phase of the SNC paradigm, animals exposed to MIA on GD15-16 and GD18-19 reach remarkably higher rates of consumption in comparison to those exposed on GD10-11 or saline. While deficits in anticipatory pleasure have been found in individuals diagnosed with schizophrenia, consummatory pleasure is considered to be normal (Gard et al., 2007). Consummatory behaviour and its suppression following the unexpected reduction of a reward in the SNC paradigm is hypothesised to reflect a negative affective state (Barnes et al., 2014; Phelps et al., 2015). The anterior cingulate cortex also plays a role in impulse control and emotion (Decety & Jackson, 2004; Jackson, Brunet, Meltzoff, & Decety, 2006); a dopaminergic disturbance within this area may offer an explanation to the unpredicted differences in consummatory pleasure observed in the SNC paradigm. Further research identifying this possible MIA neurodevelopmental disturbance, the timing of gestational sensitivity, and its biological and phenotypic effects, will allow for an optimal paradigm for inducing the negative symptoms of schizophrenia.

The present study demonstrated that exposure to an immune response on GD15-16 and GD18-19 produced reward anticipation deficits in adult offspring, similar to those seen in schizophrenia. Unfortunately, behaviour was only assessed after PND60. As schizophrenia typically develops post-puberty, behavioural analyses performed earlier in life (e.g. approximately PND35) would allow for further validation of the MIA model. Additionally, the current study did not test the efficacy of the LPS treatment, so cannot confirm whether an immune response took place or not. An extension on the current research should include either a measure of heart rate variability or cytokine presence as a result of LPS exposure. The

assessment of other MIA induced negative symptoms (such as effort-based responding as a test of motivation), and positive symptoms of schizophrenia are of interest using the current MIA protocol. Validation of the MIA model of schizophrenia offers a significant opportunity to provide a better understanding of the aetiological mechanisms that contribute in the development of this debilitating and widespread disorder; allowing for the informed development of efficacious prevention and treatment strategies for diverse schizophrenia symptomology.

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